

Enabling light-driven 3D control of nanomotors with optical forces

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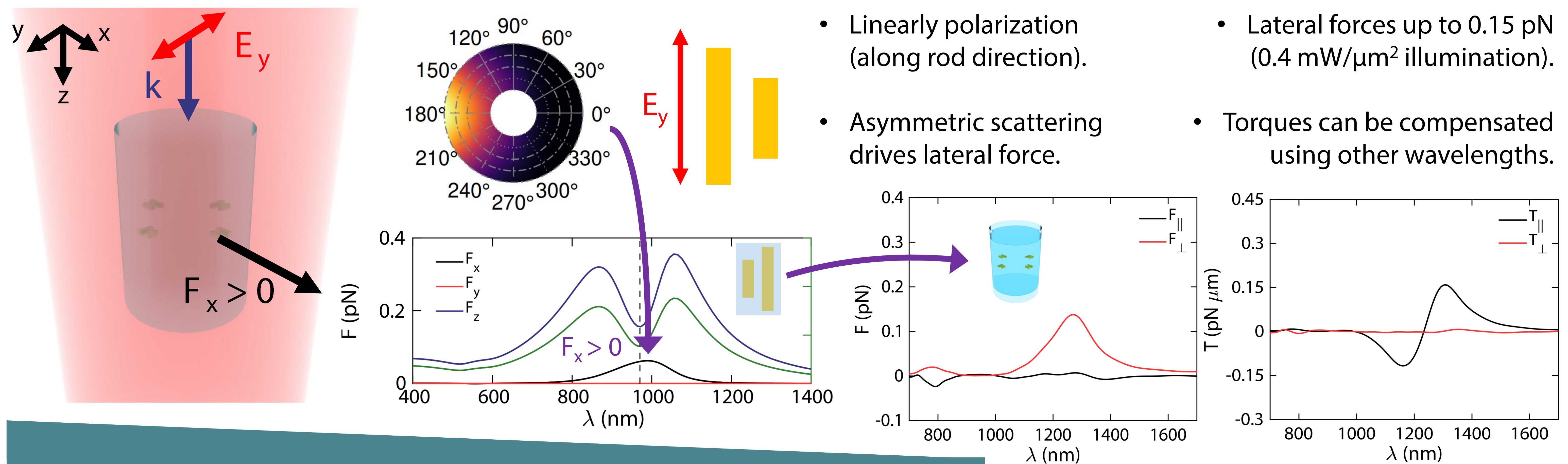
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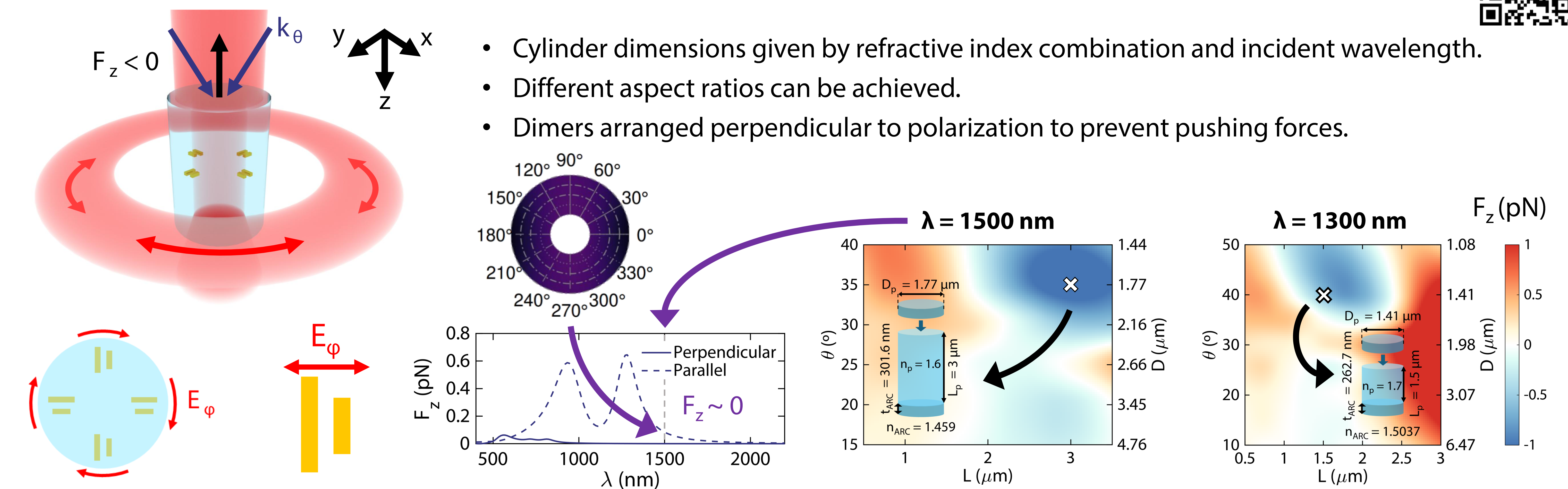
Abstract: Recent advances in nanophotonics have used scattering forces from plasmonic nanoantennas or dielectric metasurfaces to achieve transverse two-dimensional motion [1-3]. However, controlling longitudinal motion remains challenging. We introduce a nanomotor design enabling both transverse and longitudinal control: optical pulling forces from an azimuthally polarized Bessel beam act on a glass cylinder [4], while asymmetric plasmonic dimers drive lateral motion under plane-wave illumination.

Lateral movement powered by asymmetric scattering



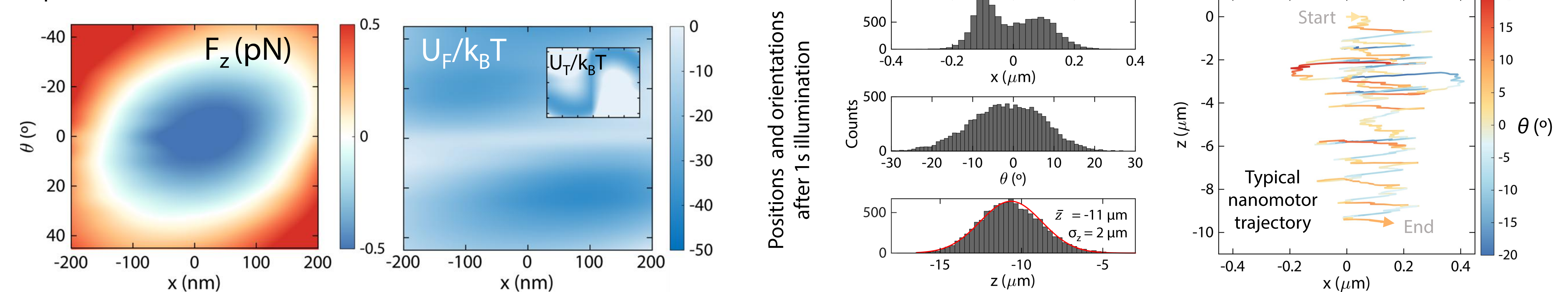
Optical pulling by an azimuthally polarized Bessel beam

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Optical pulling stability and diffusion simulations

- Optical pulling is stable to displacements and rotations.
- Non-trivial dynamics: mismatch between force and torque potential wells → Need for diffusion simulations.
- Cross-talk between torques and transverse forces make the nanomotor travel mainly through the optical pulling region → Stable pulling for long times.



References

- [1] Y. Y. Tanaka et al., *Sci. Adv.*, **6**, 45 (2020).
- [2] X. Wu et al., *Nat. Nanotech.* **17**, 5 (2022).
- [3] D. Ardrén et al., *Nat. Nanotech.* **16**, 9 (2021).
- [4] X. Li et al., *Sci. Adv.*, **5**, 3 (2019).

Conclusions

- Our design allows for independent control of transversal and longitudinal motion.
- Despite the non-trivial forces and torques, optical pulling is stable for long illumination times.
- This design can be further optimized or extended with other structures/materials.

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